

Validation Match Rate by Source Month(with +- 1 tolerance)

Share of records where validation = match; includes an overall All Months result.

Source Month	Match %	Match Count	Total Records	Diverge Category	Count	Share
2024-07	94.1%	127	135	dormant	165	38.4%
2024-08	44.4%	4	9	active	156	36.3%
2024-09	97.7%	2,431	2,487	dying	109	25.3%
2024-10	93.3%	678	727			
2024-12	96.9%	2,557	2,638			
2025-03	74.5%	102	137			
2025-05	75.8%	116	153			
2026-01	94.3%	2,609	2,768			
All Months	95.3%	8,624	9,054			

Divergence Root-Cause Analysis — engineer vs engine

Rule verified from data: a record is "match" only when engineer and engine agree within ±1 unit on ALL THREE fields (max, rop, min). This absolute ±1 rule reproduces column K for all 9,054 rows with zero mismatches. Note: the 10% relative tolerance named in the title above does NOT reproduce column K (it mislabels 1,462 rows).

Population check

Metric	Records	Share	Note
Total records	9054		1 All data rows
Diverge	430	0.047492821	Breaches ±1 on at least one field
Match	8624	0.952507179	Within ±1 on all three fields

LEVEL 1 — Where divergence concentrates

1a. Diverge rate by lifecycle category

Category	Total	Diverge	Diverge %	% of all diverges
active	257	156	0.607003891	0.362790698
dying	454	109	0.240088106	0.253488372
dormant	8343	165	0.019777059	0.38372093
Total	9054	430	0.047492821	1

1b. Diverge rate by source month

Source Month	Total	Diverge	Diverge %	% of all diverges
2024-07	135	8	0.059259259	0.018604651
2024-08	9	5	0.555555556	0.011627907
2024-09	2487	56	0.022517089	0.130232558
2024-10	727	49	0.067400275	0.113953488
2024-12	2638	81	0.03070508	0.188372093
2025-03	137	35	0.255474453	0.081395349
2025-05	153	37	0.241830065	0.086046512
2026-01	2768	159	0.057442197	0.369767442
Total	9054	430	0.047492821	1

1c. Diverge % — category x month cross-tab

Source Month	active	dying	dormant
2024-07	0.583333333	0.111111111	0
2024-08	0.5	1 n/a	
2024-09	0.675675676	0.26984127	0.005865103
2024-10	0.536585366	0.261538462	0.01610306
2024-12	0.56	0.228070175	0.0109135
2025-03	0.823529412	0.352941176	0.104651163
2025-05	0.625	0.23255814	0.139534884
2026-01	0.602941176	0.2	0.036116505

LEVEL 2 — Which field breaks, and who is higher

2a. Which field breaches the ±1 tolerance (diverge rows only)

Breaching field(s)	Diverges	% of diverges
MAX	138	0.320930233
MAX+ROP	90	0.209302326
MAX+ROP+MIN	132	0.306976744
MAX+MIN	24	0.055813953
ROP	3	0.006976744
ROP+MIN	9	0.020930233
MIN	34	0.079069767
Total	430	1

2b. Field involvement (fields overlap, so this sums >100%)

Field	Diverges involving it	% of diverges
MAX	384	0.893023256
ROP	234	0.544186047
MIN	199	0.462790698

2c. Direction of disagreement, split by category

Direction	active	dying	dormant	Total	% of diverges
engineer higher	46	26	165	237	0.551162791
engine higher	96	75	0	171	0.397674419
mixed	14	8	0	22	0.051162791
Total	156	109	165	430	1

LEVEL 3 — How big is the disagreement

3a. Gap-magnitude bands (worst absolute gap across max/rop/min)

Gap band	Diverges	% of diverges	active	dying	dormant
2 (marginal)	201	0.46744186	58	48	95
3-5 (moderate)	145	0.337209302	64	28	53
6-10 (large)	34	0.079069767	17	11	6
11-50 (severe)	33	0.076744186	9	18	6
51+ (extreme)	17	0.039534884	8	4	5
Total	430	1	156	109	165

3b. Gap severity statistics (diverge rows only)

Statistic	Worst gap (units)
Median	3
Average	13.37906977
90th percentile	12
Maximum	1200
Diverges with gap >10	50

3c. Zero-transition type on MAX (structural on/off disagreement)

Transition type	Diverges	% of diverges		active	dying	dormant	
engineer 0 -> engine >		35	0.081395349		14	21	0
engineer >0 -> engine		165	0.38372093		0	0	165
both 0 on max		0	0		0	0	0
both >0		230	0.534883721		142	88	0
Total		430		1	156	109	165

LEVEL 4 — Repeat offenders and part families

4a. Recurrence profile — is divergence one-off or systematic?

Times a part diverges		Distinct parts	Diverge records	% of diverges		
1 (one-off)		154	154	0.358139535	Distinct parts that ever d	254
	2	55	110	0.255813953	Distinct parts in dataset:	3014
	3	23	69	0.160465116		
	4	15	60	0.139534884		
	5	6	30	0.069767442		
6 or more		1	7	0.01627907		
Total		254	430	1		

4b. Top 12 repeat-offender parts (ranked by diverge count)

Rank	Part Code	Description	Diverges	Worst gap	Category
1	500699364	FILTER,ASSEMBLY,DII		7	5 active / dying
2	500175684	FERRULES NYLON 1/4		5	14 active / dying
3	500833955	REAM FILE SET		5	10 active / dying
4	500699350	CBL_TOOL_FLOW_SE		5	8 active / dying
5	500771779	PLACE TOOL 33,SiSiC		5	6 active
6	500699384	AIRFILTER,LOWPRESS		5	5 active
7	500320265	PLASMA JET NOZZLE F		5	3 dying
8	500237420	SILICONE SUCTION C		4	338 active / dying
9	500804460	BARECUVERSIONOFF		4	247 active / dying
10	500041276	FITTING, QUICK DISC		4	36 active / dying
11	500841920	FUSE 500V IEC SLO 5		4	27 active / dying
12	500699905	DISPOSABLE EARPLU		4	18 active / dying

Ranked by diverge count, then worst gap (desc), then part code. 15 parts tie at 4 diverges — ranks 8-12 show the 5 with the largest worst gap. Category lists the lifecycle states seen on that part's diverge rows only.

